■ Fractions

Comparing & Ordering Unlike Fractions

Grade 3 Math Lesson

■ What Are Fractions?

A fraction shows a **part of a whole**.

- The **numerator** (top number) = how many parts we have
- The **denominator** (bottom number) = how many equal parts the whole is split into

Example: **1/2** means one part out of two equal parts — like splitting a pizza in half! ■

■ What Are Unlike Fractions?

Unlike fractions have **different denominators** (bottom numbers).

They show parts of a whole, but the whole is divided into **different numbers of pieces**!

See how the same "1 part" looks different with different denominators:

■ The more pieces the whole is divided into, the SMALLER each piece gets!

■ How to Compare Unlike Fractions

Step 1	Step 2	Step 3
Find a Common Denominator Look for a number both denominators divide into evenly.	Make Equivalent Fractions Change each fraction so they share the same denominator.	Compare Numerators The bigger top number = the bigger fraction!

✏■ Example 1: Comparing 1/2 and 1/3

Step	What We Do	Result

1	Find common denominator of 2 and 3	→ 6
2	Convert 1/2 → multiply top & bottom by 3	= 3/6
2	Convert 1/3 → multiply top & bottom by 2	= 2/6
3	Compare numerators: 3 > 2	1/2 > 1/3 ✓


1/2


1/3

■ Ordering 3 Fractions: 1/4, 1/2, 1/3

Find a **common denominator of 12** to compare all three:

Fraction	Multiply By	Equivalent	Numerator
1/4	× 3/3	3/12	3
1/2	× 6/6	6/12	6
1/3	× 4/4	4/12	4

Order: 3/12 < 4/12 < 6/12 → So: **1/4 < 1/3 < 1/2 ■**

■■ Common Mistakes to Avoid

■ WRONG	■ RIGHT
Compare numerators WITHOUT finding a common denominator first.	Always find a common denominator FIRST, then compare.
Think larger denominators = bigger fractions. (e.g., saying 1/4 > 1/3 because 4 > 3)	Convert first! 1/4 = 3/12 and 1/3 = 4/12 So 1/3 > 1/4

■ Practice Time!

Q1: Which is bigger: 2/5 or 3/10?

Hint: Common denominator is 10. 2/5 = 4/10 and 3/10 = 3/10 → 4/10 > 3/10, so **2/5 is bigger!**

Q2: Order smallest to largest: 3/8, 1/2, 5/8

Hint: Common denominator is 8. 1/2 = 4/8 → 3/8, 4/8, 5/8 → **3/8 < 1/2 < 5/8**

■ Quick Recap

■ **Find Common Denominator:** Look for the smallest number all denominators divide into evenly.

■ **Convert Fractions:** Change each fraction to an equivalent fraction with the common denominator.

■ **Compare Numerators:** The bigger top number = the bigger fraction!

✍■ **Draw to Check:** Use fraction bars or pie charts to verify your answer visually.

"Math is fun when we take it one step at a time!" ■